

Lesson Plan 2

Descriptive data

Area of learning: Computer Science / Artificial Intelligence Literacy

Grade level: 9 (Preparatory year)

Duration: 40 minutes

Topic outline: Representation in Artificial Intelligence: Designing UI through Prompts

Outline:

1. Introduction to Representation in AI (with brief prior-knowledge check)
2. Discussion on how AI represents an idea (text → code → visual)
3. Prompt-writing activity: V1 (vague) vs V2 (structured 5-part recipe)
4. Counter-prompt extension (if time permits)
5. Reflection and student questions
6. Exit ticket (in class, ~2 min, silent, hand in at the door)
7. Summative assessment (assigned as homework)

Instructional objectives :

By the end of the lesson, students will be able to:

- **Explain the concept of Representation in AI systems.**
 - AI4K12 BI2 (Key Insight) - CSTA 3A-AP-13 - MoNE/MEB TYMM: BTYAB2
- **Identify the three layers of representation involved in generating a webpage with AI: natural-language prompt, HTML/CSS code, and rendered visual page.**
 - AI4K12 BI 2 - AI4K12 2-A-i.9-12 (introduced through analogy) - ISTE: Standard 5
- **Construct a structured UI prompt using the 5-part recipe (Audience, Purpose, Content, Style, Structure).**
 - ISTE Standard 6 - MoNE/MEB TYMM: BTYAB1.2 - CSTA 3A-AP-15
- **Demonstrate that a more specific prompt produces a more accurate output by comparing two AI-generated webpages built from the same goal.**
 - ISTE Standard 5 - MoNE/MEB TYMM: Eleştirel Düşünme
- **Recognize that an AI's default style and layout choices come from patterns in its training data, and describe one limitation of AI-generated UI for a context relevant to the student.**
 - AI4K12 3-A-i.9-12 - AI4K12 5-A-i.9-12 - CSTA 3A-IC-24 - MoNE/MEB TYMM: Eleştirel Düşünme; Dijital Okuryazarlık

Standards Alignment Summary

- AI4K12 Big Idea 2 – Representation & Reasoning (Primary)
 - "Computers construct representations using data structures; representations support reasoning, and reasoning methods operate on representations."

- 2-A-i.9-12: "Describe how AI agents represent knowledge using internal data structures (schemas, frames, semantic networks)." (*Introduced via schema analogy; foundational preparation for full LO coverage in later modules.*)
- AI4K12 Big Idea 3 – Learning
 - 3-A-i.9-12 (Learning from Data): "Describe how machine-learning models acquire patterns from training data."
- AI4K12 Big Idea 5 – Societal Impact
 - 5-A-i.9-12 (Bias in Data and Models): "Describe how bias in datasets can affect AI behaviour and outputs."
- CSTA (2017) Level 3A:
 - 3A-AP-13: "Describe how AI drives many software and physical systems."
 - 3A-AP-15: "Justify the selection of specific control structures and data representations."
 - 3A-IC-24: "Evaluate the ways computing impacts personal, ethical, legal, and cultural practices."
- **MoNE/MEB TYMM:** BTYAB1.2 (Bilişim Teknolojileri ile Çalışma) · BTYAB2 (Algoritmik Düşünme) · Dijital Okuryazarlık · Eleştirel Düşünme
- **ISTE:** Standard 5 (Computational Thinker), Standard 6 (Creative Communicator)

Other materials

- Devices with internet access
- Free online tools, no sign-up required: a generative AI assistant (ChatGPT or Google Gemini) and CodePen.io as the live HTML/CSS preview environment.
- [Appendix A:](#) Student worksheet & Answer Key
- [Appendix B:](#) 5-part prompt recipe card (*student handout*)
- [Appendix C:](#) "Same prompt, two outputs" demo screenshots. (*For teacher*)
- [Appendix D:](#) Three pre-generated webpages (*For teacher*)
- [Appendix E:](#) Summative Assessment Quiz & Answer Key

Detailed explanations of the lesson:

Steps	Explain what the teacher does /says	Explain what the students are expected to do	Related Standards & Conceptual Connections
1 10 mins (5mins intro + 5 mins setup)	Introduction: The teacher begins with a 1-minute prior-knowledge check before showing the demo. Teacher: "Quick question before we start, has anyone here used an AI tool like ChatGPT, Gemini, or something similar? What did you ask it to do? Just shout out a few examples." Teacher takes 3–4 short answers from around the room (no writing yet). The goal is to surface what students already think AI is for	Students briefly call out what they have used AI tools for (no writing). Typical answers: homework, writing essays, answering questions, summarising things. Students sit ready with their device and worksheet. They observe the two example webpages on the board and note one	AI4K12 Big Idea 2 – Representation & Reasoning (introduced) ISTE Standard 5 – Computational Thinker

	Teacher: "Good. Today, we are going to use it to build something you can see. Look at these two websites." The teacher begins by projecting two AI-generated webpages on the smart board. Both were generated from prompts about a small Istanbul café, but the prompts were very different in detail. Students see the visual surprise: two completely different-looking websites for the "same" idea. Teacher: "Look at these two websites. Which café would you actually want to visit? Which one looks more like a real Istanbul café? They were both made by the same AI -ChatGPT -in less than ten seconds. So why are they so different?" "The answer is in what I told the AI. For the first one, I just wrote: 'make a website for a café.' For the second one, I described who comes to the café, what it sells, what mood I wanted, and how the page should be organized." "What actually happens when you write a prompt? Your words go in, but a webpage comes out. In the middle, the AI is doing something, it's building an internal picture of what you mean, and then turning that picture into HTML and CSS code, and then your browser turns that code into a visual page. So one idea, three different forms. We call this representation, and that's what we are exploring today." "Your goal today isn't just to make a pretty website. Your goal is to learn how to describe an idea so clearly that the AI can build what you actually imagined. By the end of class, you will write two prompts for the same goal and see, with your own eyes, why writing a better prompt is itself a skill. Let's start." Teacher distributes the worksheet (Appendix A) and the 5-part prompt recipe card (Appendix B). Setup: Students open two tabs — a generative AI assistant (ChatGPT or Gemini) and codepen.io. Most have a Google or ChatGPT account; those who don't pair with someone who does. Teacher has 2 backup accounts.	visible difference in worksheet Section A. Expected responses to "Why are they so different?": "The first one is more detailed." (partial correct but doesn't name the cause) "You ran it twice and AI is random." (misconception – the difference is the prompt, not randomness) "The second prompt told the AI more, so the AI built a clearer picture of what you wanted." (target) During the setup phase, students open two browser tabs (AI assistant + CodePen) and confirm they are logged in. Pairs without a personal account share a partner's device.	
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2 22 mins (10+ 12) mins	Main activity: Mini-lesson (10 min): Teacher draws the 3-layer diagram: natural language → HTML/CSS → rendered page. "The AI doesn't go from your words straight to the picture. It first builds a structured idea in code. The cleaner your description, the cleaner the code, the closer the page will be to what you imagined." Teacher introduces the 5-part recipe (HAİSY) on the board: A — Audience (Hedef Kitle) P — Purpose (Amaç) C — Content (İçerik) S — Style (Stil) S — Structure (Yapı) "This recipe is a kind of schema a structured way to represent your idea. AI uses schemas internally too: a Restaurant schema has slots for name, location, cuisine, hours. You're doing the same thing for a webpage. If you skip a slot, the AI fills it with defaults learned from its training data we'll come back to this in the closure." Activity 1 – V1 vs V2 (12 min): Pairs choose one project: school club page / local business / school event / personal portfolio. "For your project, you will write two prompts for the same goal. V1 is one short, vague sentence. V2 uses the full 5-part recipe. Run both in your AI assistant, copy each piece of HTML+CSS into a fresh CodePen tab, and screenshot both results. Then mark in worksheet Section B5 the two specific differences you can see between V1 and V2." Teacher circulates with one-on-one help and the prompt: "Which slot gave the AI the most useful information? Which was hardest to fill?" No whole-class talk pair conversations work better with prep-year students.	Mini-lesson: Students label the 3 layers of representation on the worksheet diagram (natural language / code / visual) and copy the 5-part recipe into Section B-2 of the worksheet. Main Activity 1 – V1 vs V2: In pairs, students choose a project from the list and write both V1 and V2 prompts on the worksheet (Section B-5). They run V1 in the AI assistant, paste the HTML+CSS into a CodePen tab, screenshot the result, and repeat for V2. They mark two specific differences between the V1 and V2 outputs in Section B-5 (checkbox + category format). Expected differences students might write (Section B-3): •"V2 has more sections." (partial – true but surface-level) •"V1 is just a blue page with the word Café. V2 has a header, a menu, a contact section." (target – specific and observable) •"V2 is better because the prompt was longer." (misconception – length isn't the cause; structured information is.)	AI4K12 Big Idea 2 -Representation & Reasoning Key Insight: representations support reasoning; reasoning methods operate on representations. 2-A-i.9-12 (introduced through schema analogy) CSTA 3A-AP-15 – Justifying selection of representations ISTE Standard 6 – Creative Communicator MoNE/MEB TYMM: BTYAB1.2 – Bilişim Teknolojileri ile Çalışma PRIMM: Predict (V2 will be better), Run (both prompts), Investigate (compare outputs)
3	Extension activity: <i>If the class has worked through V1 vs V2 calmly and</i>	Students go to the "opsiyonel" box, find	AI4K12 Big Idea 5 - Societal Impact

<i>Optional - 10 mins</i>	<i>there is time before closure, the teacher introduces this short extension. If pacing is tight, the step is skipped — LO 5 is still anchored through the closure reflection and the quiz. Teacher points students to the small “OPSİYONEL” box at the bottom of the worksheet.</i> Teacher: "Look at your V2 output again. Find ONE thing the AI decided on its own that doesn't fit your project — a colour, an image, the language, a section. Write a short refinement prompt that overrides only that one thing." Example: "If the AI gave you a blue café and you wanted Turkish-inspired colours: 'Change the colour palette to terracotta and cream tones, keep everything else the same.'" Students run the refinement prompt, paste the new HTML+CSS into a third CodePen tab, and check whether only the targeted element changed. Teacher asks one or two pairs: "Did only the thing you wanted change, or did the AI change other things too?" This builds awareness for the closure on AI defaults.	ONE element in their V2 output that the AI chose on its own and doesn't fit (colour, image, language, or section), write a short refinement prompt targeting only that element, run it, and paste the result into a new CodePen tab. They tick one checkbox: "Only the thing I wanted changed" (well-targeted) or "Something else changed too" (prompt too broad -coaching opportunity). Expected target answers in the OPSİYONEL box: <i>Targeted change:</i> "Change the colour palette to terracotta and cream tones, keep everything else the same." <i>Cultural override:</i> "Rewrite the entire page in Turkish, do not change the layout."	5-A-i.9-12 Bias in Data and Models (experienced through override) AI4K12 Big Idea 2 -Representation & Reasoning (reinforced) ISTE Standard 6 -Creative Communicator MoNE/MEB TYMM: Eleştirel Düşünme
4 <i>3-4 mins</i>	Closure activity: Teacher pauses and brings the class together for a brief reflection — two questions, each with a short reframing afterwards, and then deliberately opens the floor for student questions before the silent exit ticket. Reflection question 1: "Someone in this room thought V2 was better because the prompt was longer. Is that the real reason?" After giving students 30 seconds to answer (in pairs or to the whole class), the teacher reframes: "Length alone isn't the reason. A long prompt with no structure is just noise. What helped V2 was the organisation audience, purpose, content, style,	Students briefly share their thoughts in response to the reflection questions — to a partner first, and then optionally to the class. Expected responses to question 1 ("was V2 better just because it was longer?"): •"Yes, longer is better." (misconception) •"It was more detailed." (partial) •"It wasn't the length, it was that the information was organised: audience, purpose,	AI4K12 Big Idea 2 (re-stated through reflection 1) AI4K12 Big Idea 3 - Learning 3-A-i.9-12 Learning from Data AI4K12 Big Idea 5 - Societal Impact 5-A-i.9-12 Bias in Data and Models CSTA 3A-IC-24 – Computing's impact on cultural practices MoNE/MEB TYMM: Eleştirel Düşünme

	structure. The AI built a cleaner internal representation because you gave it cleaner inputs." Teacher quietly switches the smart board to the three pre-generated webpages from Appendix D (bakery / wedding planner / tech startup), so the visuals are on display while the question is asked. The pages stay in the background as students answer they make the AI's default style choices concrete instead of abstract. Reflection question 2: "When you didn't tell the AI what colour or layout to use, it filled in defaults on its own. Whose websites did the AI learn those defaults from? What does it not know about us in Istanbul?" After hearing two or three answers, teacher reframes: "The AI learned from websites on the internet mostly English, mostly American or European, mostly Western design. So when you ask for a 'modern café' with no detail, you get a Silicon-Valley café, not a kahvehane. It isn't picking from templates inside the program; it's generating from learned patterns. Knowing this makes you a designer with the AI, not just a user of it." Teacher gives a one-sentence recap: "Today you saw that the same goal can produce two very different webpages, and the difference was how clearly you represented your idea using the 5-part recipe." "Any quick questions before the exit ticket?" Teacher takes 1–2 student questions, answering briefly. If a question is too big for the remaining time, teacher notes it on the board for the next lesson. Teacher distributes Appendix E and announces silent assessment time. "For wrapping up this lesson, I want you to answer this part of the document. The rest will be your homework to complete at home. or we will start with it as a warm-up next lesson."	content, style, structure." (target) Expected responses to question 2 ("whose websites did the AI learn from?"): •"From Google." (partial) •"From websites on the internet." (partial) •"Mostly Western, English-language websites so it doesn't know our local style / kahvehane / Turkish weddings." (target) Students ask any remaining quick questions. Students work silently for ~2 minutes, writing three sentences in their own words on the exit-ticket slip, and hand it in as they leave. They take the rest of the portion of Appendix E home.	
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5 2 mins (Exit ticket)	Summative assessment (AppendixE): Only the quiz portion of Appendix E (multiple choice, fill-in, matching items 1–6) is administered outside the 40-minute window. The exit ticket portion (items 7–9, three sentences in students' own words) is completed in class during Step 4. The teacher hands the quiz portion to students at the very end of Step 4 with one of two options, at the teacher's discretion: •Option A: Students complete the quiz at home and submit at the start of the next lesson. •Option B: The teacher opens the next lesson with the quiz as a 5-minute silent warm-up, then collects it before introducing new material. <i>This split preserves the full 5-LO coverage of the original assessment: the in-class exit ticket gives a same-day formative read on whether the closure landed, and the take-home quiz tests recall and application with a beat of reflection.</i> Quiz items 1–6 + exit-ticket items 7–9 map to LOs 1–5. Formative assessment is used throughout the lesson: pair-work engagement, V1 and V2 CodePen renders, worksheet sections A–B (kept by student), oral answers during closure.	Students take the quiz portion of Appendix E with them at the end of class. They complete it at home (Option A) or silently at the start of the next lesson (Option B) and hand it in. Note: students are assessed only on their written work and their CodePen outputs, not on what they said aloud which is appropriate for prep-year learners who are still building confidence to speak about new technical topics.	Aligned to all 5 instructional objectives. AI4K12 Big Ideas 2 (primary), 3, 5 Formative + summative assessment alignment
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Suggestions for teachers who may implement this lesson plan in the future:

1. Detailed explanation of how technological tools will be used:

Students use a generative AI assistant (ChatGPT or Google Gemini) to turn structured prompts into HTML/CSS code, then paste the code into CodePen.io to see the rendered webpage instantly. Both tools are free, browser-based, and require no account creation — essential for a 40-minute class of 25 prep-year students.

2. Are there any cautions or points to consider about the technological tools that are integrated into the lessons?

- Avoid tools that require registration. Platforms such as Figma Make, v0, Uizard, or Galileo AI were intentionally avoided because creating accounts can waste valuable class time.

- AI outputs may vary. Generative AI tools may produce different results from the same prompt. This should be presented as part of the lesson, showing that AI systems do not always respond in identical ways.
- Code may contain errors. Sometimes the generated HTML/CSS may not work perfectly. This can be used as a teaching opportunity to discuss the limits of AI tools.
- Internet connection may slow down. If many students use AI tools at the same time, the network may become slower. If necessary, students can work in pairs.
- Students may become distracted. Although unlikely with UI design tasks, students should be asked to stay within the worksheet topics. The teacher should monitor screens during the activity.
- Preparation is recommended. Teachers should test sample prompts on ChatGPT and Gemini before the lesson to understand possible outputs and prepare examples.

3. Other suggestions:

Lesson Structure and Differentiation

This lesson is intentionally calm-paced and focuses mainly on the V1 vs V2 comparison activity, which addresses most of the learning objectives. The counter-prompt extension is treated as truly optional only run if pacing allows. The summative assessment has been split: the exit ticket is completed silently in class at the end of Step 4, while the quiz items are taken home or used as the warm-up for the next lesson. The plan totals 37 minutes of core instruction with ~3 minutes of buffer for transitions and login issues.

Worksheet Use

Teachers are encouraged to distribute the worksheet on paper rather than digitally so students do not need to switch between multiple tabs. Including one completed example in the 5-part prompt template can also help students understand what a successful prompt looks like before writing their own. Students should be encouraged to save a screenshot of their best V2 result, as this can increase motivation and support later lessons on reading and writing HTML/CSS.

Classroom Management

- Use a single-page worksheet so students can see the full activity without feeling overwhelmed.
- Write the current section (A / B) on the board and update it throughout the lesson.
- Use quick pair check-ins to identify students who may need support.
- Reduce long writing tasks by using checkboxes and short-answer formats.
- Include a short setup period for opening browser tabs and confirming logins.
- Give optional extension tasks only to pairs who finish early.
- Encourage students who fall behind to focus on completing the V2 activity first.
- Use the final minutes for the reflection, a brief student-questions moment, and the silent exit ticket; only the quiz portion goes home.
- Begin with a short reassurance that the worksheet is a guide for learning, not an exam.